

Activity: maximum likelihood estimation for GLMs

GLMs

We say that a distribution belongs to the *exponential dispersion model* family if the distribution's probability function can be written in the form

$$f(y; \theta, \phi) = a(y, \phi) \exp \left\{ \frac{y\theta - \kappa(\theta)}{\phi} \right\}$$

For iid observations $(\mathbf{x}_1, Y_1), \dots, (\mathbf{x}_n, Y_n)$, a **generalized linear model** (GLM) consists of two components:

$$\begin{aligned} Y_i | \mathbf{x}_i &\sim \text{EDM}(\mu_i, \phi) \\ g(\mu_i) &= \mathbf{x}_i^T \beta \end{aligned}$$

For this activity, we will assume that g is the **canonical link function** – that is, $g(\mu_i) = \theta_i$ (the canonical parameter).

Furthermore, recall the following facts about EDMs:

- $\frac{d}{d\theta_i} \kappa(\theta_i) = \mu_i$
- $\frac{d^2}{d\theta_i^2} \kappa(\theta_i) = \frac{d\mu_i}{d\theta_i} = V(\mu_i)$

where $\mathbb{E}[Y_i | \mathbf{x}_i] = \mu_i$ and $\text{Var}(Y_i | \mathbf{x}_i) = \phi V(\mu_i)$.

Questions

Maximum likelihood estimation for GLMs does not, in general, have a closed form. Instead, iterative procedures like Newton's method are commonly used. Find expressions for:

- The score $U(\beta)$
- The Hessian $\mathbf{H}(\beta)$
- The Fisher information (for a single observation) $\mathcal{I}_1(\beta)$
- An *estimate* of the Fisher information